

FlashMind AI

Product Requirement Document (PRD)

Project Name: FlashMind AI

Tagline: Learn Faster with AI Generated Flashcards

Version: 1.0.0

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1. Executive Summary

FlashMind AI is a modern, production-ready, full-stack web application designed to help students automatically create, study, and manage study materials using Artificial Intelligence.

By combining the structural simplicity of Notion, the learning efficacy of Quizlet, and the conversational reasoning power of ChatGPT, FlashMind AI makes study preparation seamless, automated, and scientifically optimized.

- Live Vercel Deployment Link: <https://flash-mind-ai-sigma.vercel.app>

2. Target Audience & Personas

- The Busy Student (University/School): Needs to convert long lecture PDFs, slides, and raw textbook chapters into bite-sized, high-yield study cards quickly before exams.
- The Lifelong Learner: Needs to build long-term memory in professional subjects (coding, language, history) using structured, repetitive review cycles.

3. Product Features & Functional Requirements

3.1. Document & Note Ingestion

- PDF Upload & Parsing: Users can drag-and-drop or upload PDF study materials. The system reads and extracts text locally.
- Plain Text Notes: A paste-area allowing users to dump text notes directly.
- Topic-Based Generation: Users can type a search prompt or subject topic (e.g., "Mitochondria cellular respiration") to generate cards from scratch.

3.2. Generative AI Engine

- DeepSeek & Llama Inference: Powered by high-speed NVIDIA NIM endpoints.
- Model Fallback Loop: To handle NVIDIA 503 rate limits or overloads, the server handles requests through a robust fallback chain:

DeepSeek-V3/Flash → Llama-3.1-Nemotron-70b → Llama-3.1-8b-Instruct

- Safe JSON Extraction: A custom parser isolates JSON arrays [...] or objects {...} from conversational AI prefixes/suffixes.

3.3. Subject & Deck Management

- Subject Directories: Decks are grouped under subjects (e.g., "Biology", "Computer Science") to keep workspaces clean.
- Custom Decks: Flashcards are structured into individual collections (decks) for targeted study sessions.

3.4. Scientific Spaced Repetition Study Mode

- SM-2 Algorithm Integration: Review schedules are calculated dynamically using the SuperMemo-2 (SM-2) spaced repetition algorithm:
 - *Quality Responses (0-5)*: Users self-rate their recall.
 - *Repetitions (\$n\$)*, *Easiness Factor (\$EF\$)*, and *Intervals (\$I\$)* are saved individually per card.
 - TTS Audio Support: Text-to-speech audio pronunciation triggers for auditory learners.
 - Interactive 3D Cards: CSS 3D perspectives allow users to click-to-flip cards smoothly.

3.5. Dynamic Quiz Arena

- AI Quiz Generation: Generates graded Multiple Choice, True/False, or Fill-in-the-blank questions based on the deck's content.
- Performance Feedback: Instantly shows correct/incorrect answers, provides detailed explanations, and saves progress to database history.

3.6. Gamification & Analytics Dashboard

- XP System: Users earn $+5\text{ XP}$ for creating flashcards and $+10\text{ XP}$ for answering quiz questions correctly.
- Level Badges: Automatically ranks user proficiency levels (e.g., "Scholar Level 2") based on total XP.
- Study Streaks: Tracks consecutive active study days using database-driven streak counters.
- Progress Charts: Interactive charts showing weekly cards studied and quiz success rates.

3.7. Responsive Day/Night Themes

- Dual Theme Toggle: A header toggle switches between a vibrant dark-morphic theme and a clean, high-contrast day mode.
- Aesthetics: Smooth CSS variables transitions, glassmorphic menus, and readable text contrasts on all page elements.

4. Technical Architecture & Database Schema

4.1. Core Tech Stack

- Frontend & Server Routes: Next.js 16 (Turbopack) with React 19.
- Database & Auth: Supabase (PostgreSQL with Row Level Security).
- Inference API: NVIDIA NIM API.
- Styles: Tailwind CSS v4 with custom CSS custom property overrides.

4.2. Database Design (PostgreSQL)

The application leverages the following relational database tables:

- `profiles``: Custom user level, XP, study streak tracker, and user avatars.
 - `subjects``: Subjects created by users.
 - `decks``: Decks linked to subjects.
 - `flashcards``: Individual cards holding front text, back text, and SM-2 metadata (``reps``, ``interval``, ``easiness_factor``, ``next_review``).
 - `study_logs``: History logs tracking cards studied for chart metrics.
 - `quiz_history``: Score records of completed quizzes.
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5. Non-Functional Requirements & Performance

- Lazy Loading Pagination: Grid views load a maximum of 12 complex 3D cards at once to prevent browser lag.

A "Load More" action handles pagination.

- PDF Ingestion Stability: Heavy worker processes are loaded via local bundles to prevent third-party CDN worker loading blockers.

- Safety & Security: All queries strictly obey Supabase Row Level Security (RLS) policies based on the active authenticated user ID.

